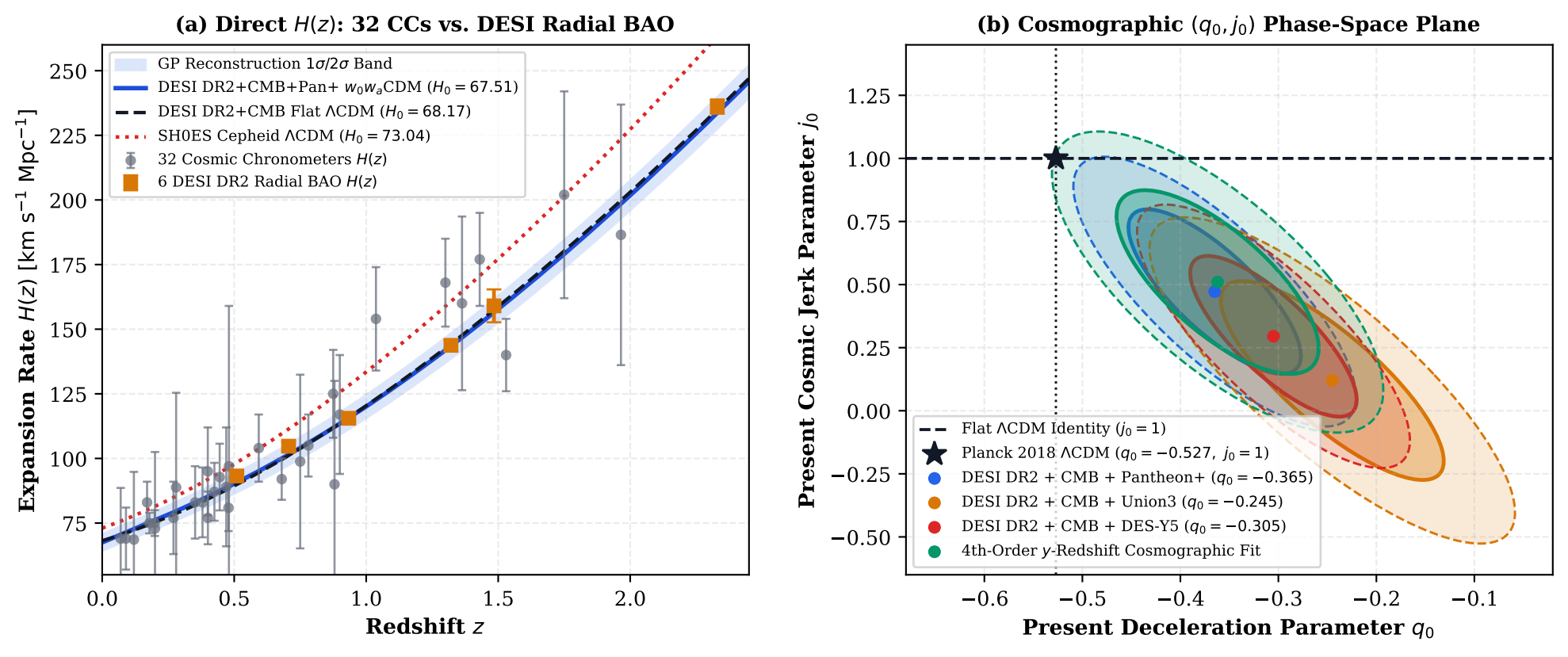

(a) Direct $H(z)$: 32 CCs vs. DESI Radial BAO
GP Reconstruction $1\sigma/2\sigma$ Band
DESI DR2+CMB+Pan+ w_0w_aCDM ($H_0 = 67.51$)
DESI DR2+CMB Flat ΛCDM ($H_0 = 68.17$)
SH0ES Cepheid ΛCDM ($H_0 = 73.04$)
32 Cosmic Chronometers $H(z)$
6 DESI DR2 Radial BAO $H(z)$
Expansion Rate $H(z)$ [km s^{-1} Mpc^{-1}]
Redshift z
(b) Cosmographic (q_0, j_0) Phase-Space Plane
Present Cosmic Jerk Parameter j_0
Present Deceleration Parameter q_0
Flat ΛCDM Identity ($j_0 = 1$)
Planck 2018 ΛCDM ($q_0 = -0.527,\ j_0 = 1$)
DESI DR2 + CMB + Pantheon+ ($q_0 = -0.365$)
DESI DR2 + CMB + Union3 ($q_0 = -0.245$)
DESI DR2 + CMB + DES-Y5 ($q_0 = -0.305$)
4th-Order y-Redshift Cosmographic Fit